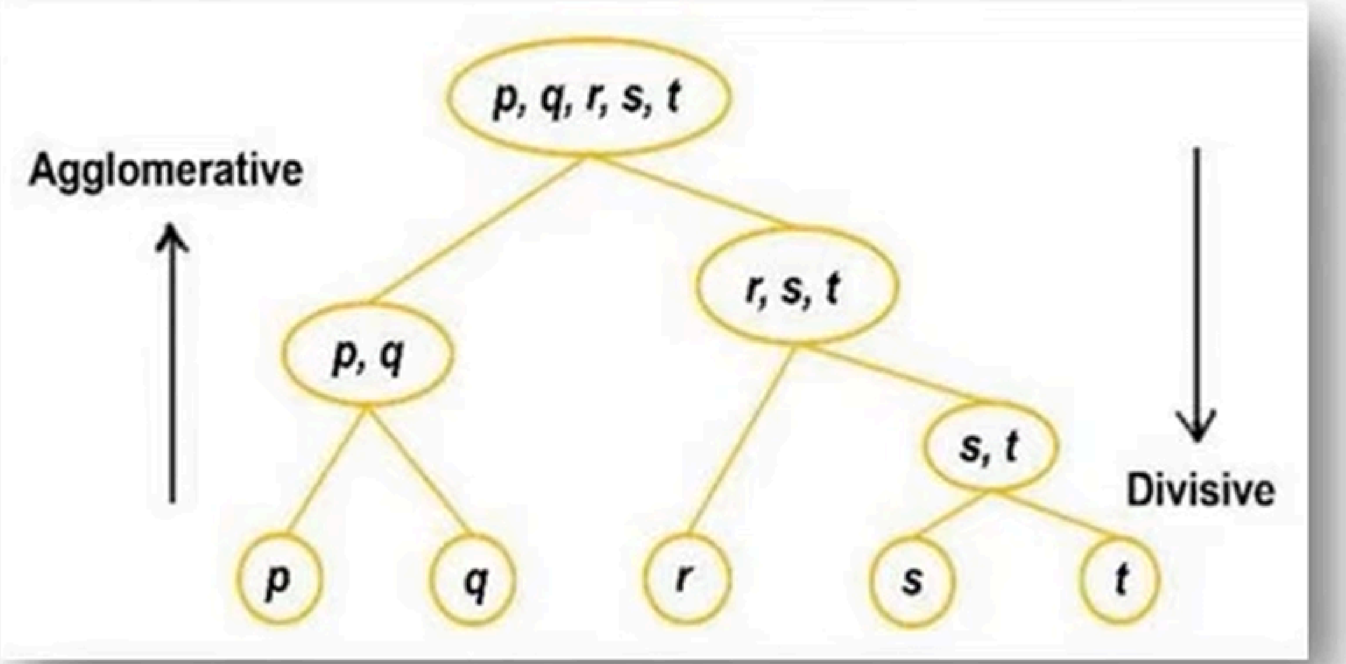
# Types of Hierarchical Clustering

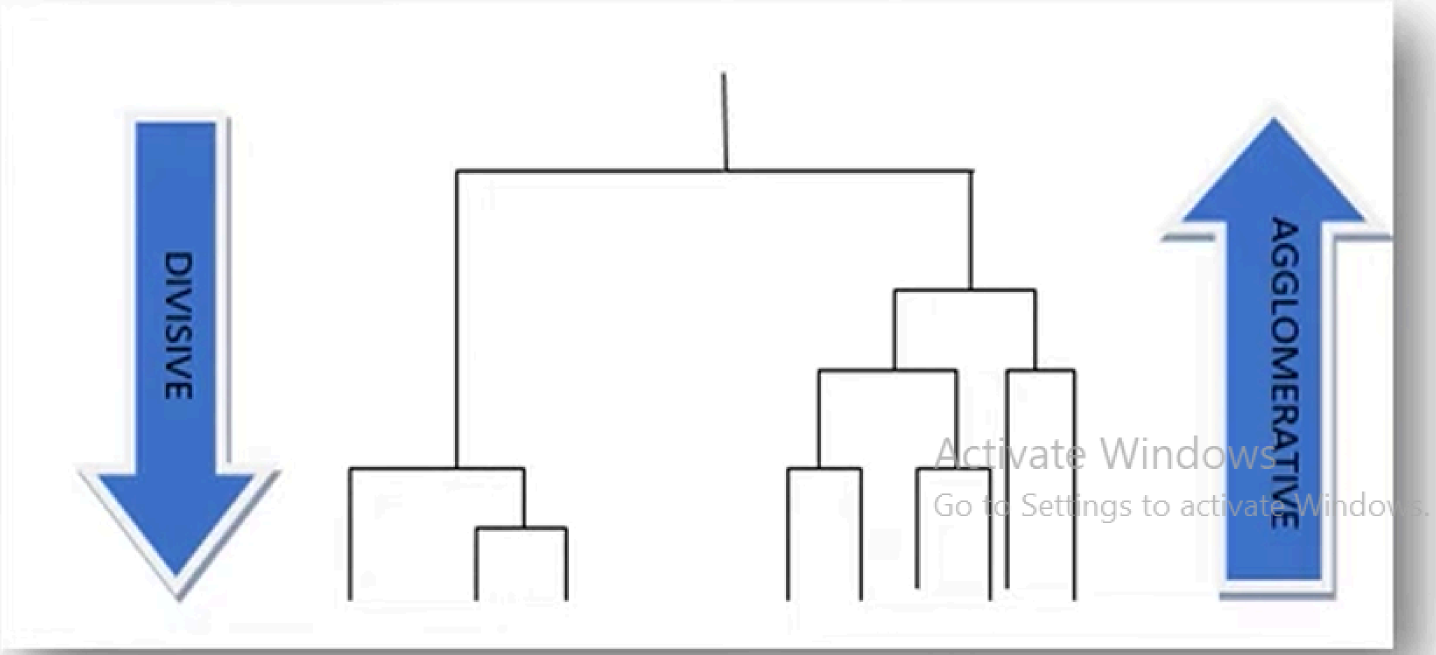
## Type 1: Agglomerative Clustering:

- Agglomerative is a **bottom-up** approach, in which the algorithm starts with taking all data points as single clusters and merging them until one cluster is left.

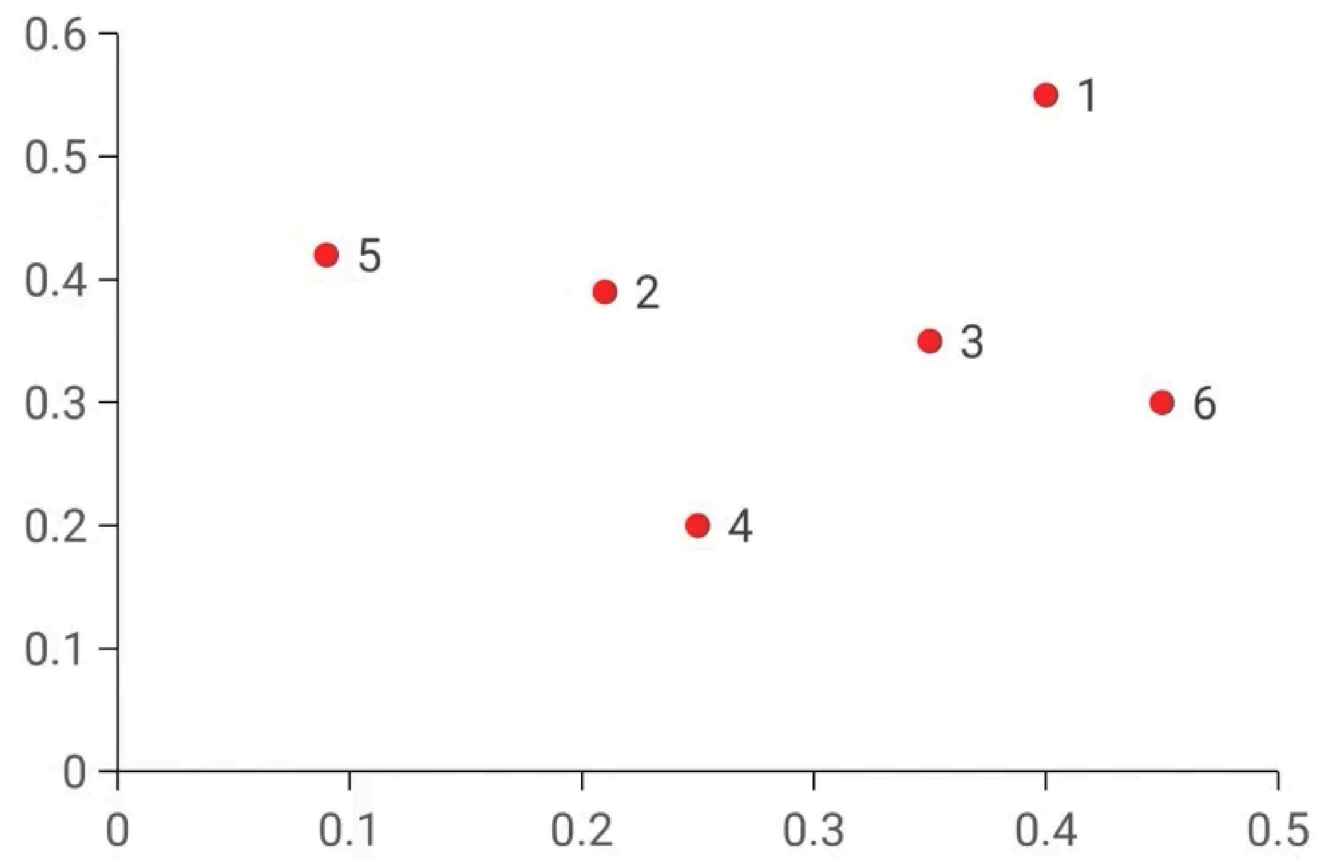
## Type 2: Divisive Clustering:

- Divisive algorithm is the reverse of the agglomerative algorithm as it is a **top-down approach.**



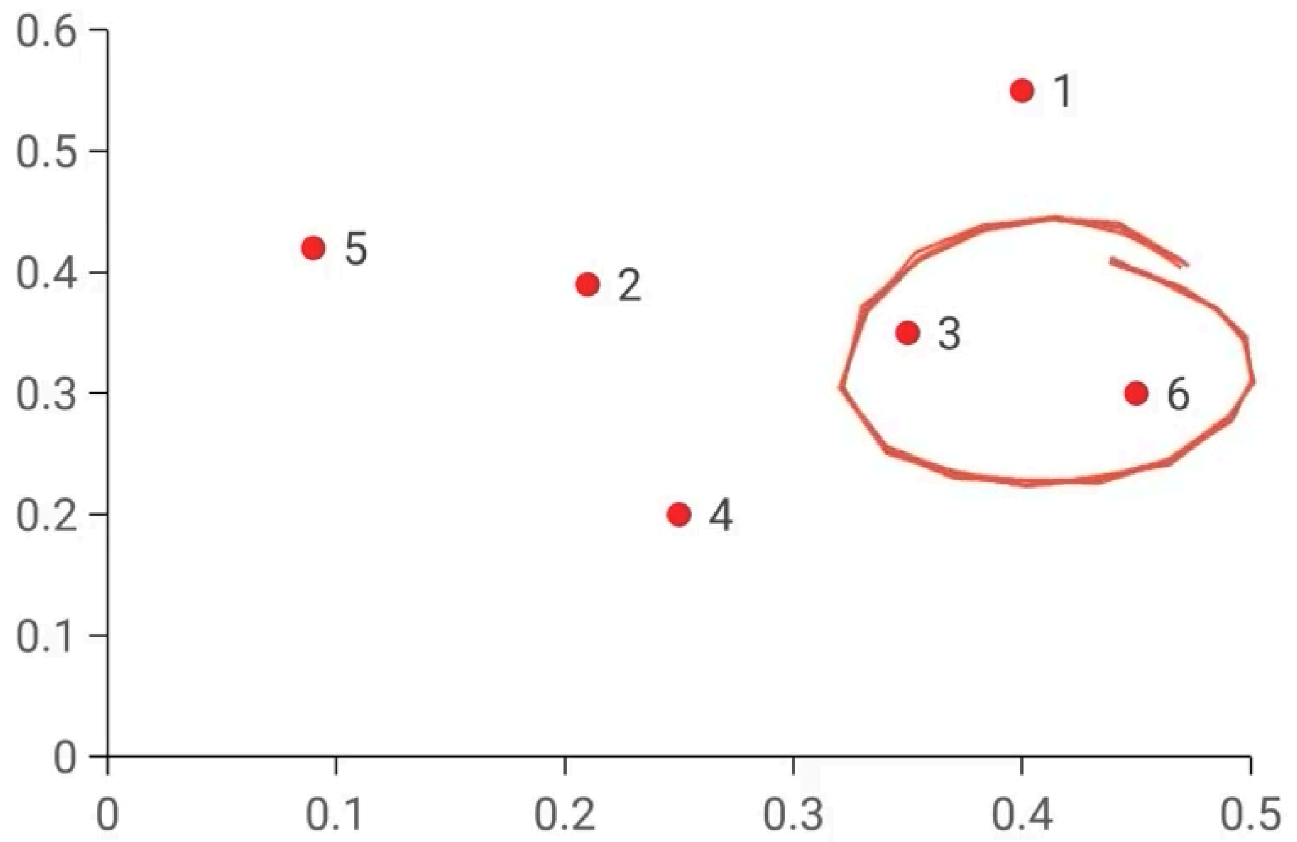
Treat each data point as one cluster




For n data points, there will be n clusters

Compute the distance between each of the data points to form a distance matrix

| | P1 | P2 | P3 | P4 | P5 | P6 |
|---|---|---|---|---|---|---|
| P1 | 0 | | | | | |
| P2 | 0.23 | 0 | | | | |
| P3 | 0.22 | 0.15 | 0 | | | |
| P4 | 0.37 | 0.20 | 0.15 | 0 | | |
| P5 | 0.34 | 0.14 | 0.28 | 0.29 | 0 | |
| P6 | 0.23 | 0.25 | 0.11 | 0.22 | 0.39 | 0 |

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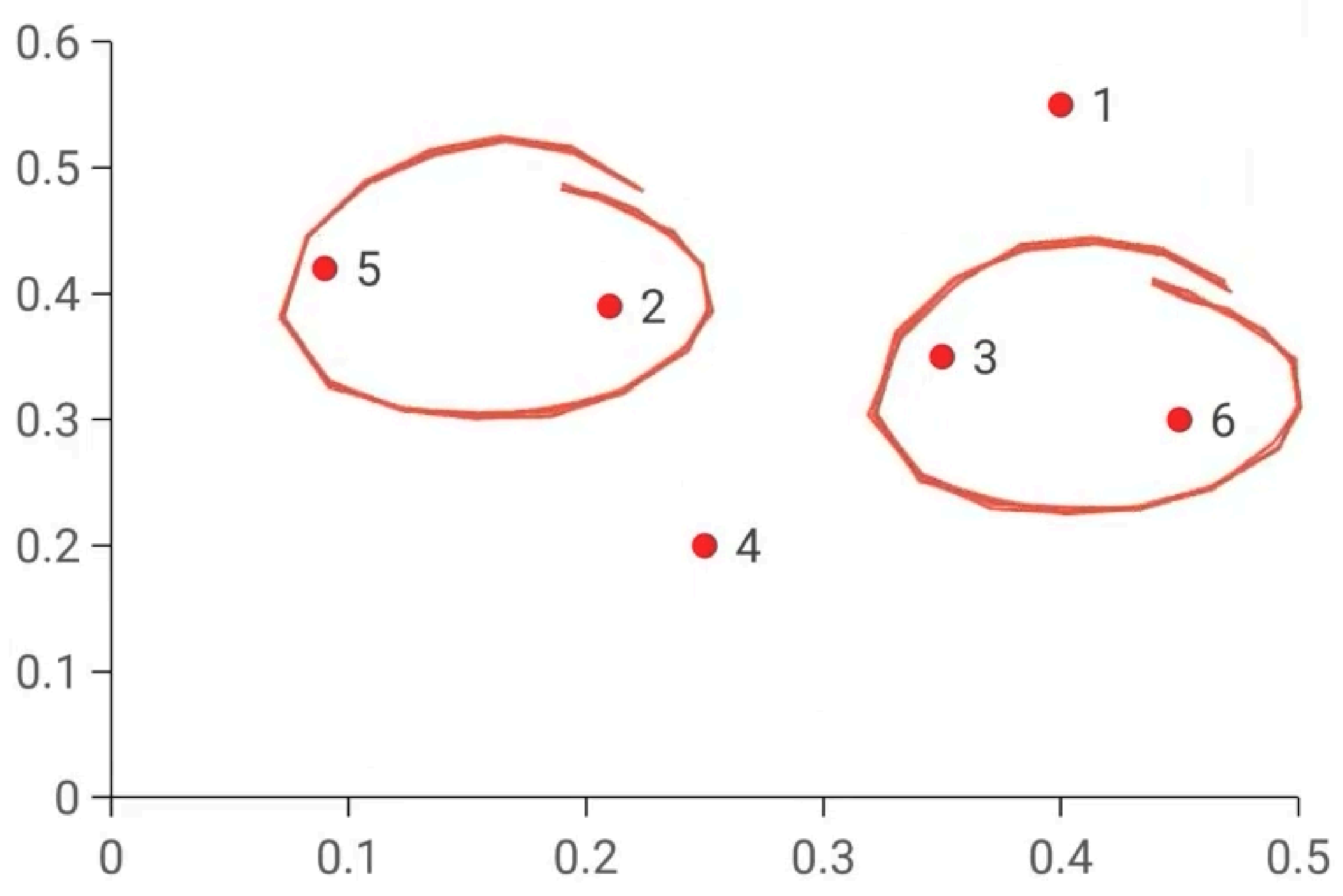
| P3 | 0.22 | 0.15 | 0 | | | |
| P4 | 0.37 | 0.20 | 0.15 | 0 | | |
| P5 | 0.34 | 0.14 | 0.28 | 0.29 | 0 | |
| P6 | 0.23 | 0.25 | 0.11 | 0.22 | 0.39 | 0 |

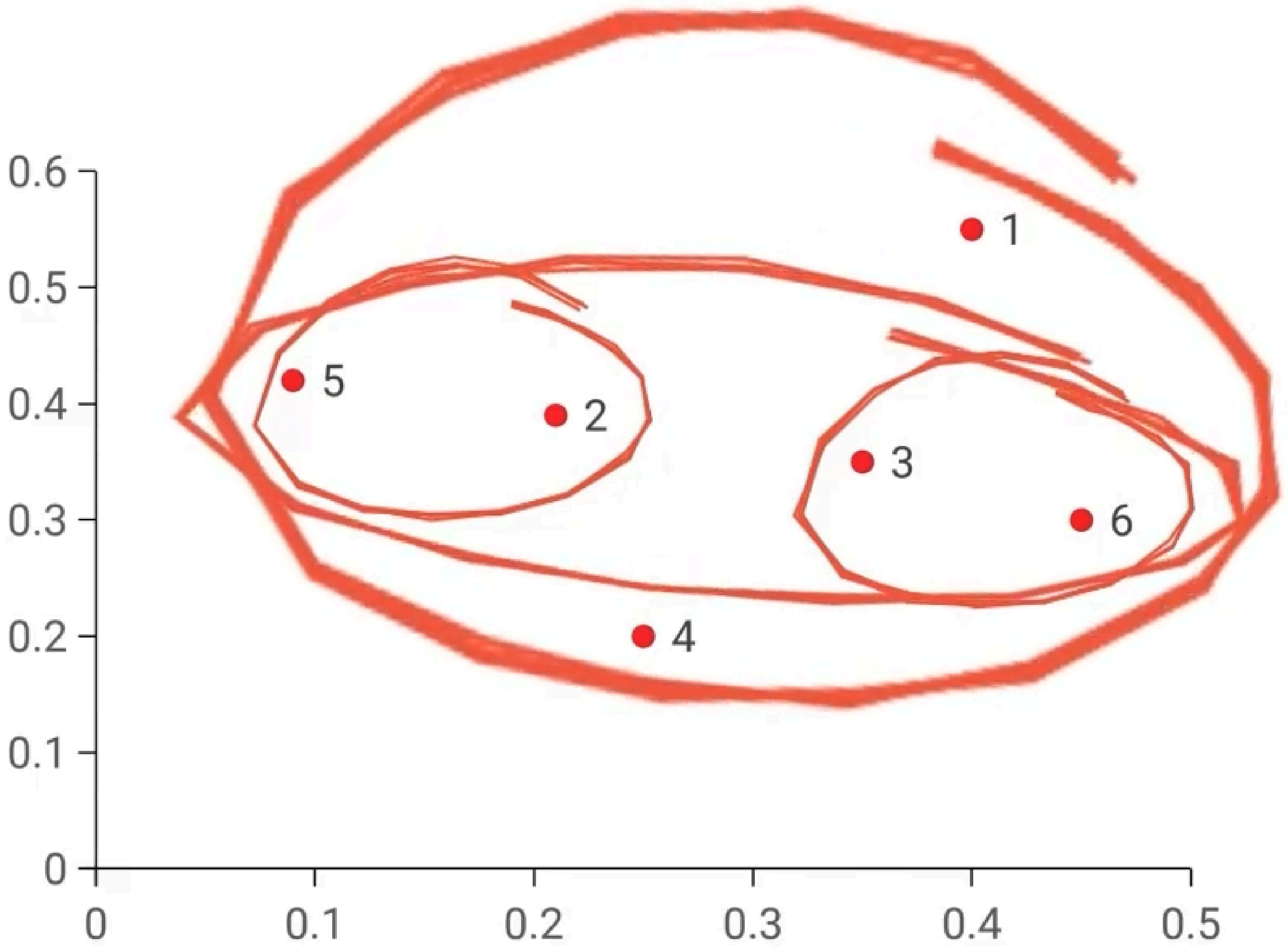
From the distance matrix find out the two closest data points

Form a cluster by joining these two data points

Combine the next two closest data points a
form a cluster by joining these two data poi



Repeat the exercise of combining the closest data points till one big cluster is formed



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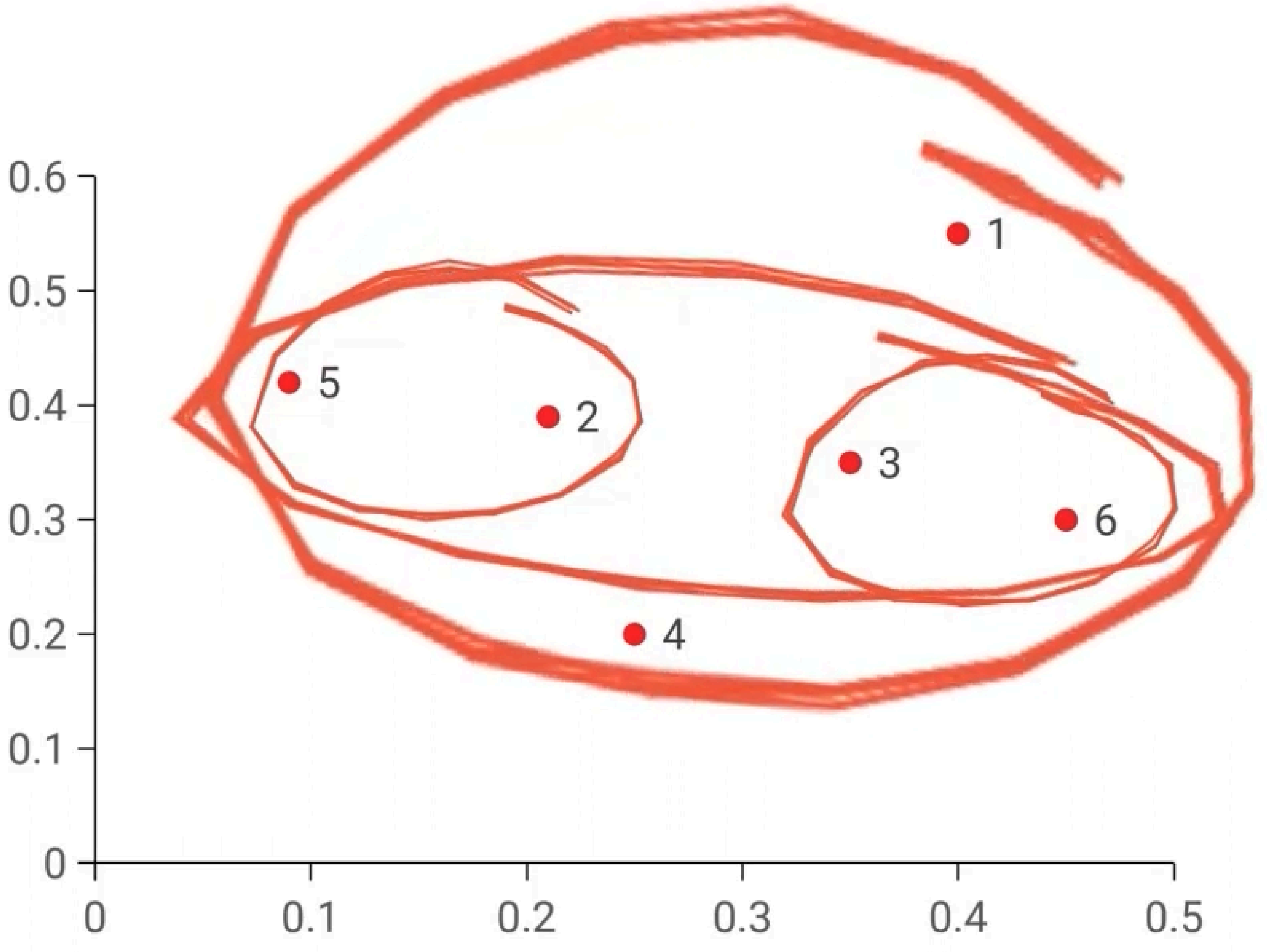
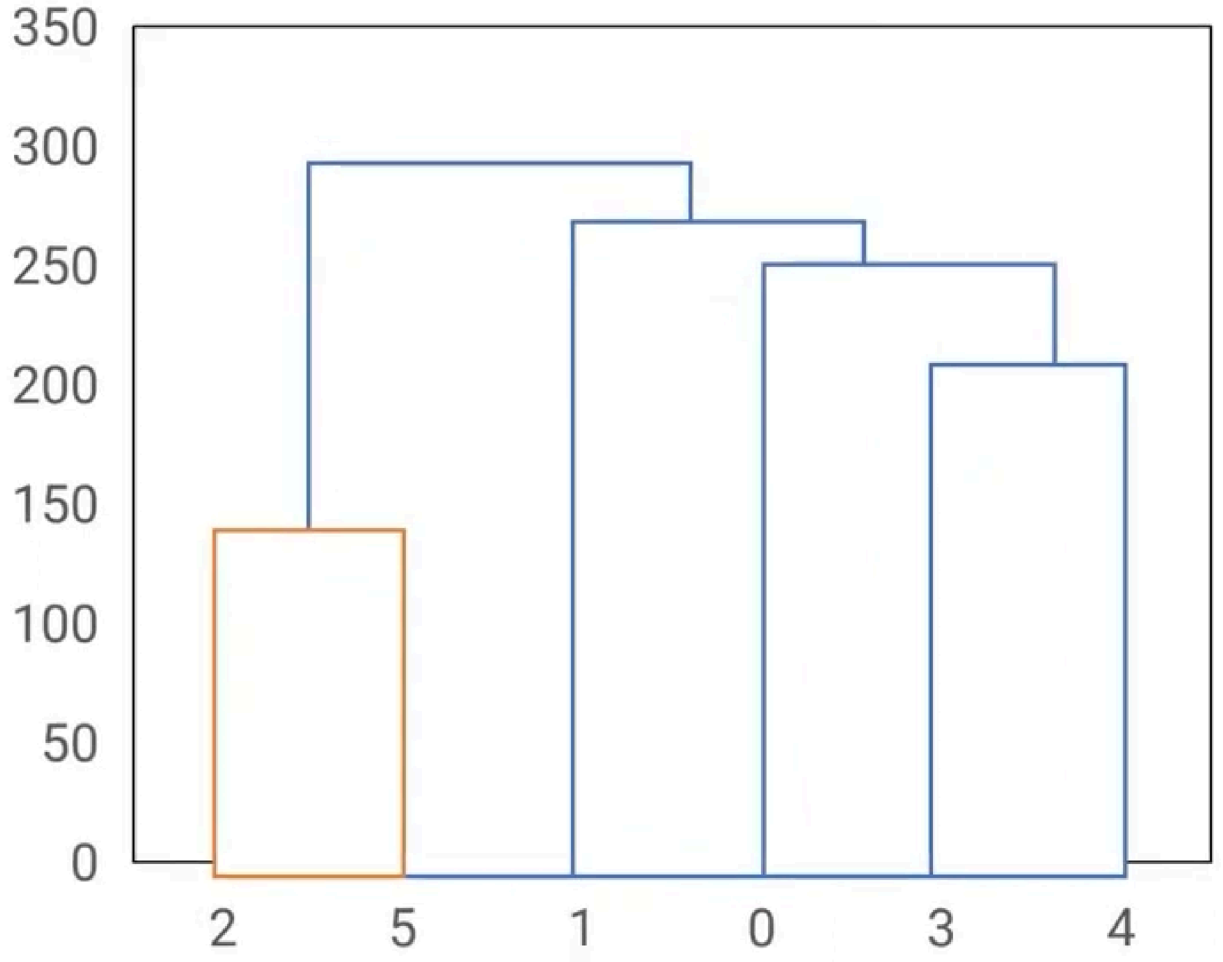



Visualize the steps through a pictorial representation - Dendrogram

To find out the distance between the newly formed cluster and the remaining data points:

**Single linkage**

Considers the shortest distance between two points in each cluster

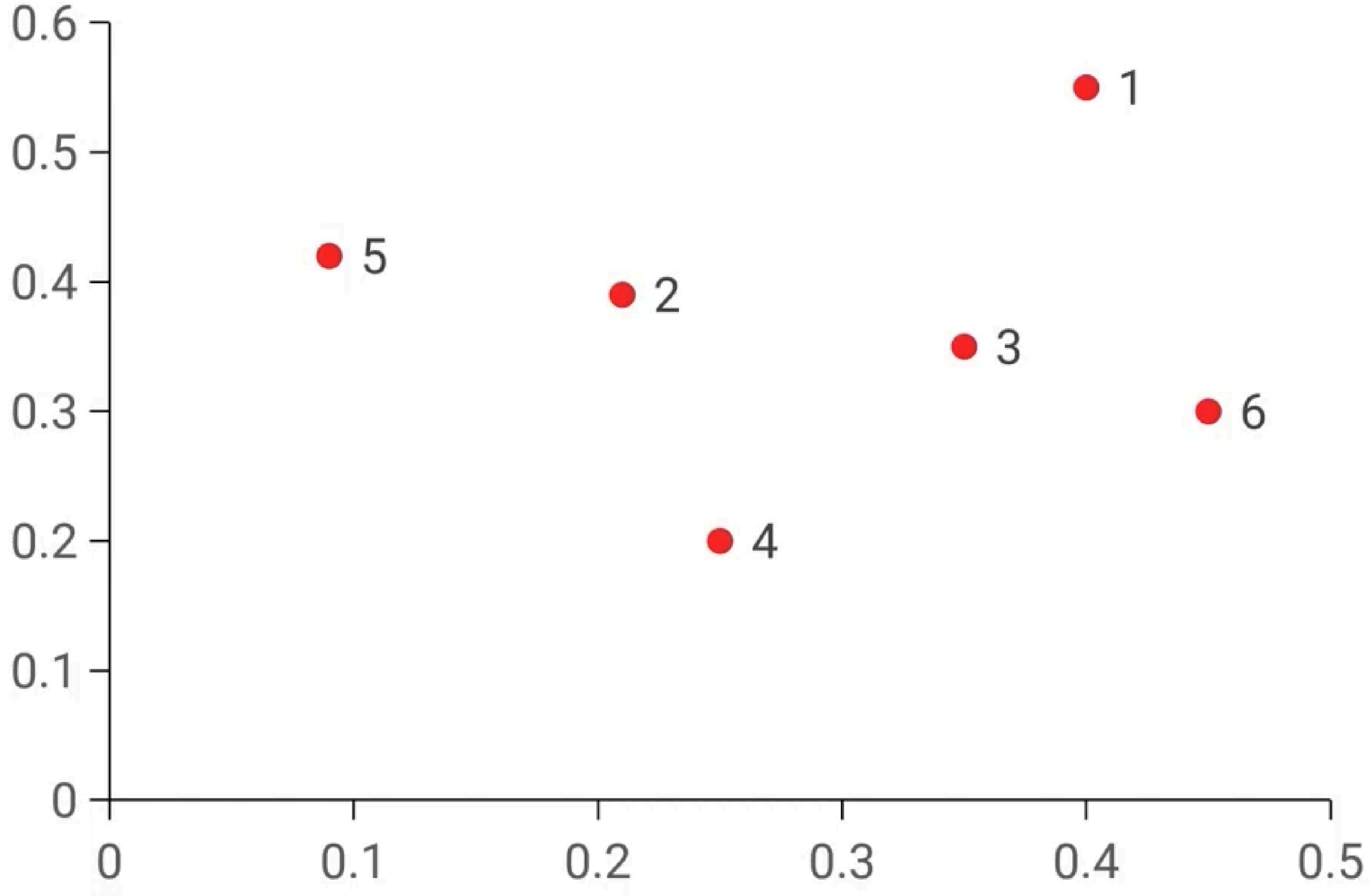
**Complete linkage**

Considers the longest distance between two points in each cluster

**Average linkage**

Considers the average distance between each point in one cluster to every point in the other cluster

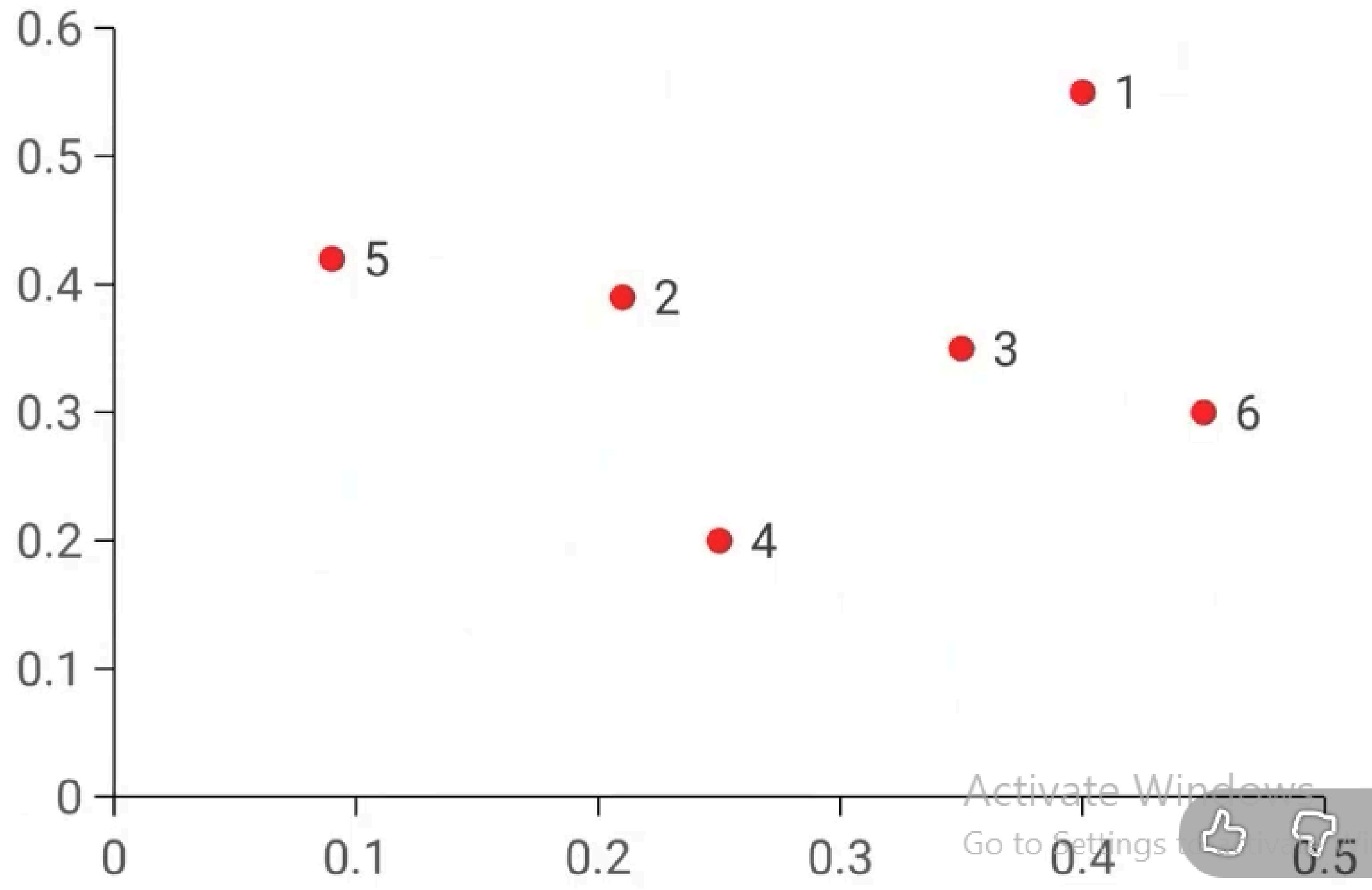
Consider the sample data points




| | X | Y |
|---|---|---|
| P1 | 0.40 | 0.53 |
| P2 | 0.22 | 0.38 |
| P3 | 0.35 | 0.32 |
| P4 | 0.26 | 0.19 |
| P5 | 0.08 | 0.41 |
| P6 | 0.45 | 0.30 |

## From the distance matrix choose the nearest two data points with minimum distance

| | P1 | P2 | P3 | P4 | P5 | P6 |
|---|---|---|---|---|---|---|
| **P1** | 0 | | | | | |
| **P2** | 0.23 | 0 | | | | |
| **P3** | 0.22 | 0.15 | 0 | | | |
| **P4** | 0.37 | 0.20 | 0.15 | 0 | | |
| **P5** | 0.34 | 0.14 | 0.28 | 0.29 | 0 | |
| **P6** | 0.23 | 0.25 | 0.11 | 0.22 | 0.39 | 0 |



Update the distance matrix using Single linkage technique

| | P1 | P2 | P3, P6 | P4 | P5 |
|---|---|---|---|---|---|
| P1 | 0 | | | | |
| P2 | 0.23 | 0 | | | |
| P3, P6 | 0.22 | 0.15 | 0 | | |
| P4 | 0.37 | 0.20 | 0.15 | 0 | |
| P5 | 0.34 | 0.14 | 0.28 | 0.29 | 0 |

To update the distance matrix
MIN [dist(P3,P6), (P1)]
MIN (dist(P3,P1), (P6,P1))
=min[(0.22,0.23)]
=0.22

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| | P1 | P2 | P3, P6 | P4 | P5 |
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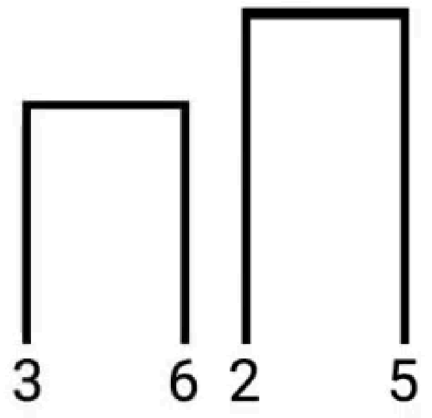
| P5 | 0.34 | 0.14 | 0.28 | 0.29 | 0 |

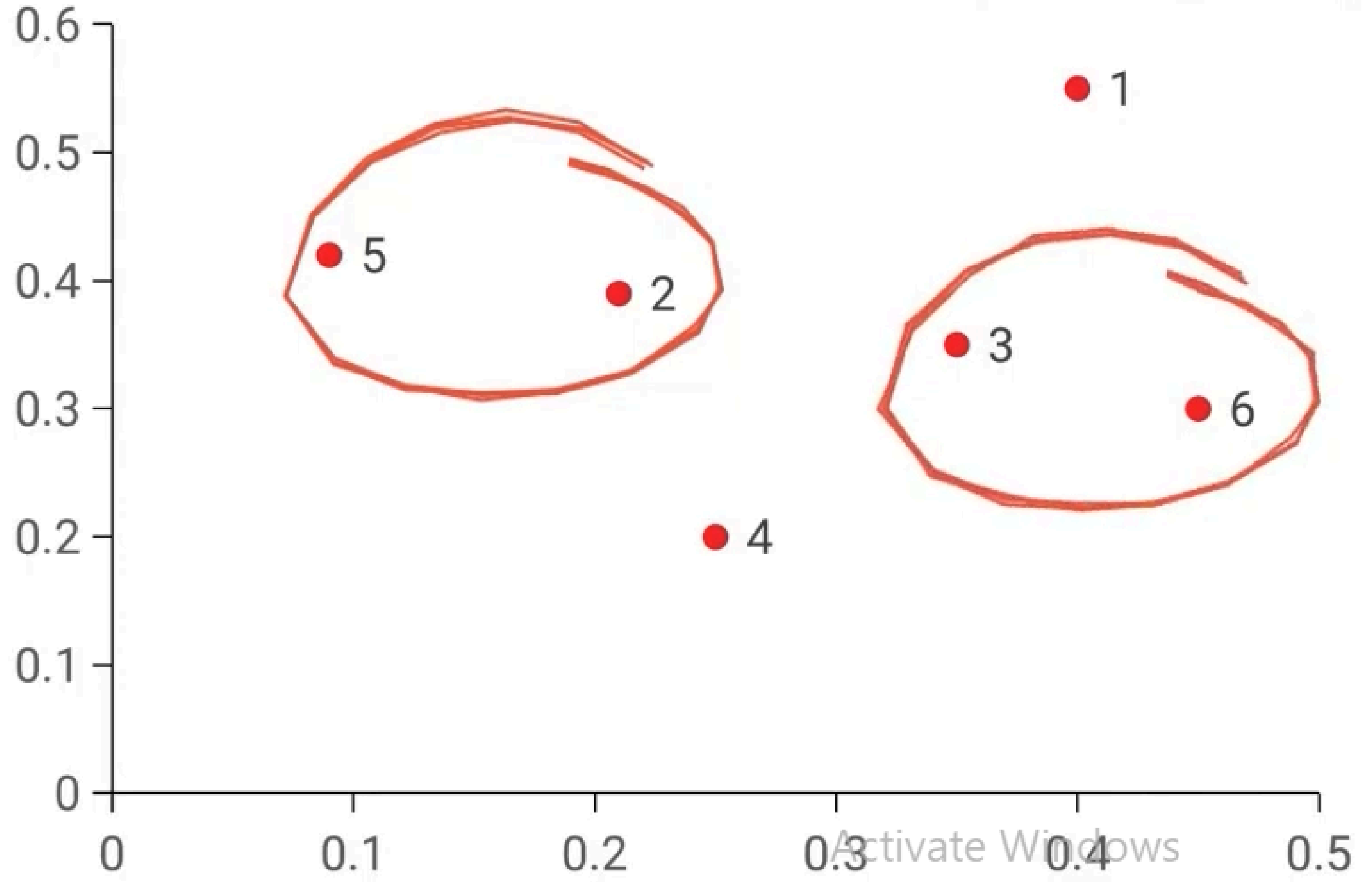
To update the distance matrix
MIN [dist(P3,P6), (P1)]
MIN (dist(P3,P1), (P6,P1))
=min[(0.22,0.23)]
=0.22

To update the distance matrix
MIN [dist(P3,P6), (P2)]
MIN (dist(P3,P2), (P6,P2))

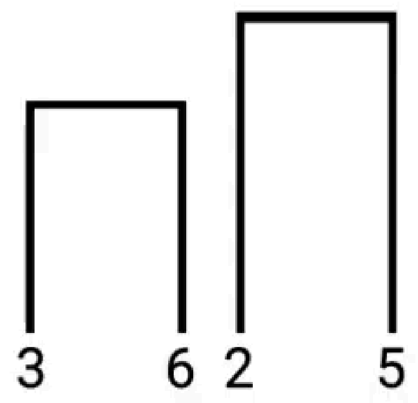
From the updated distance matrix combine the next two nearest data points to form another cluster

| | P1 | P2 | P3, P6 | P4 | P5 |
|---|---|---|---|---|---|
| **P1** | 0 | | | | |
| **P2** | 0.23 | 0 | | | |
| **P3, P6** | 0.22 | 0.15 | 0 | | |
| **P4** | 0.37 | 0.20 | 0.15 | 0 | |
| **P5** | 0.34 | $\boxed{0.14}$ | 0.28 | 0.29 | 0 |

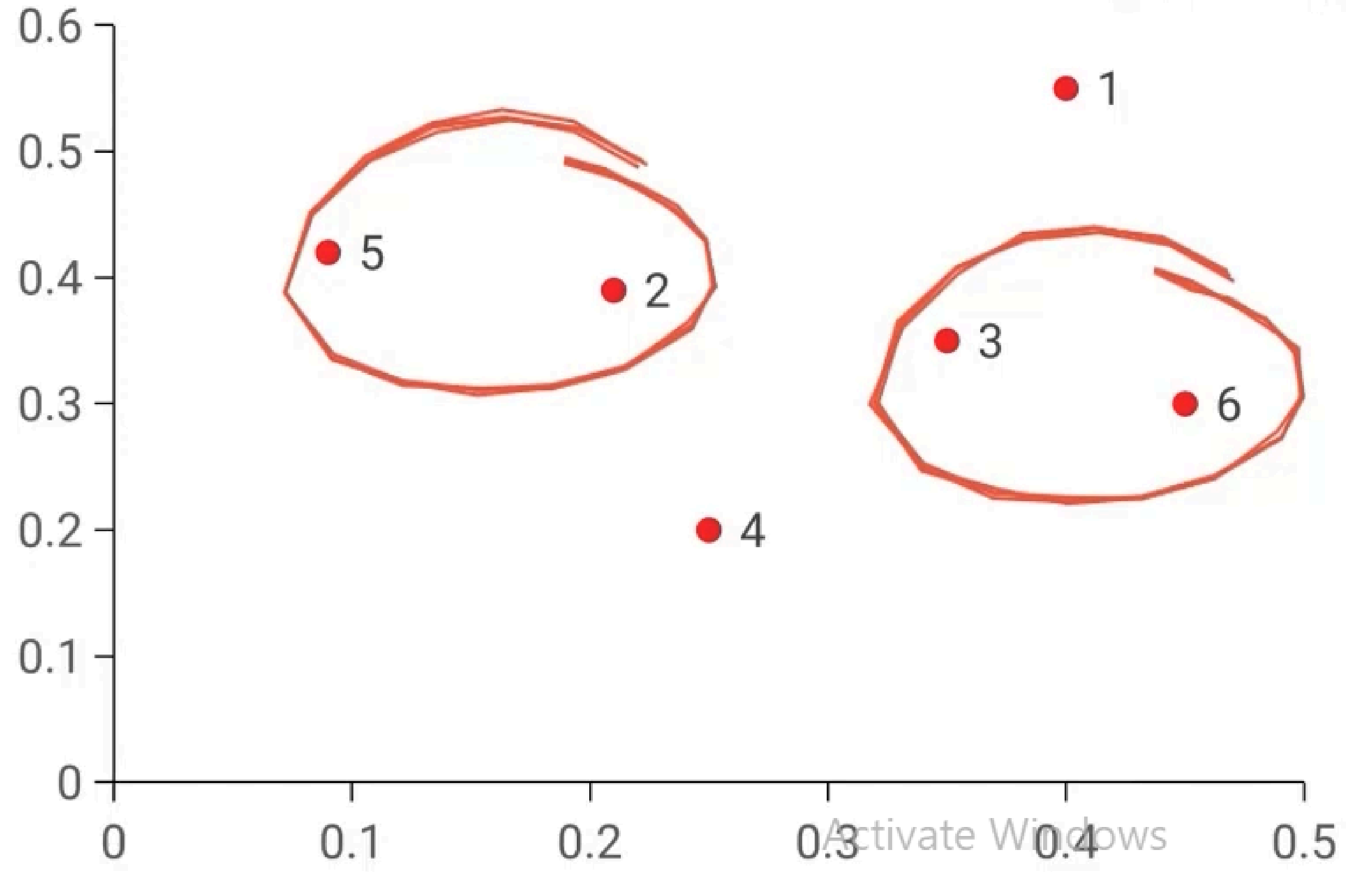


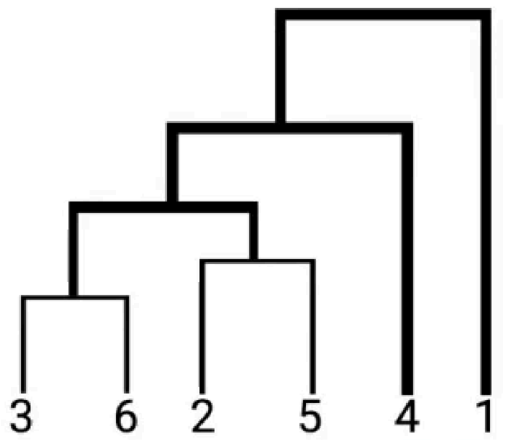
Repeat the exercise of combining the data points and updating the distance matrix, till all the data points are grouped to form one single cluster



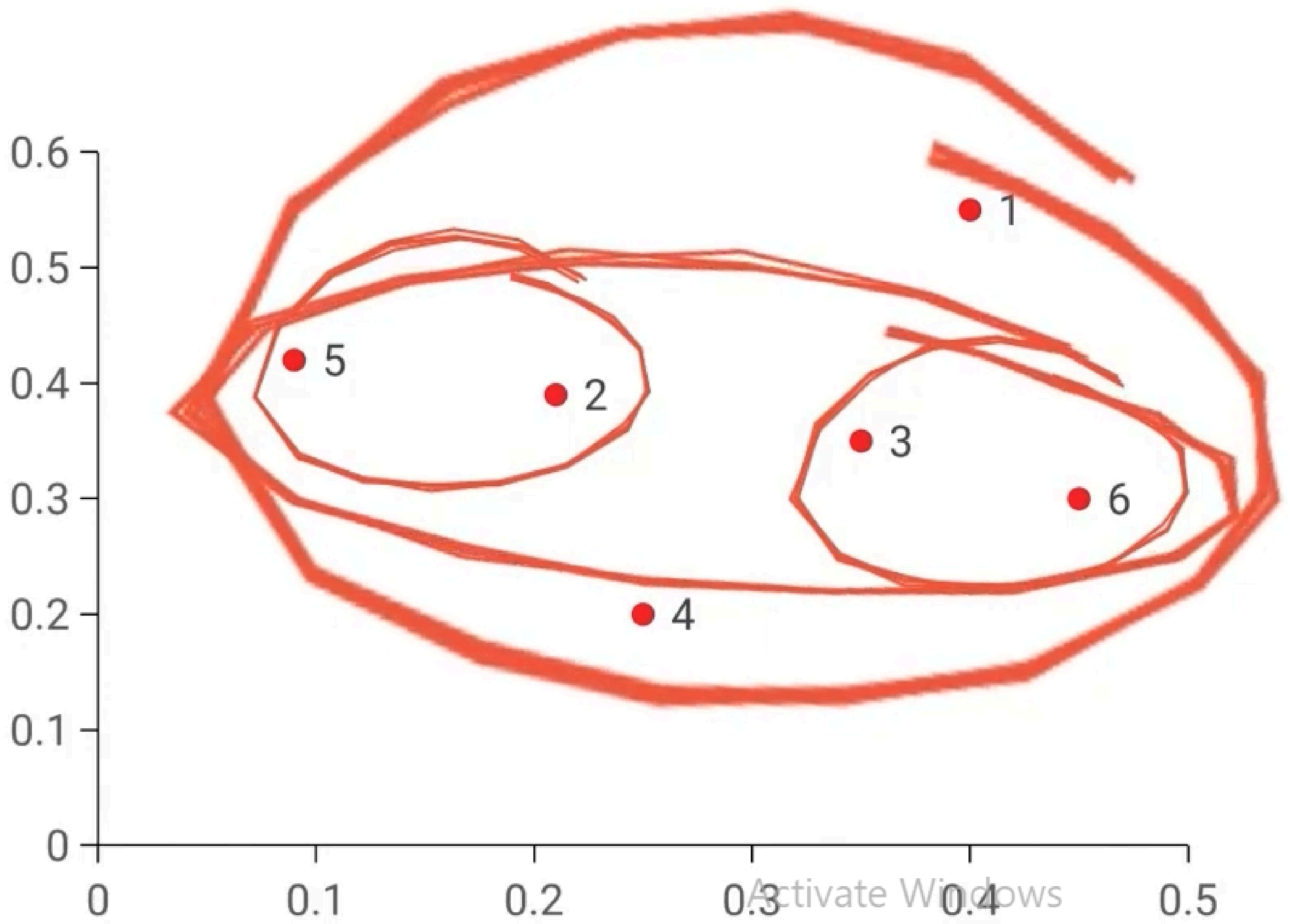

| | P1 | P2, P5, P3, P6, P4 |
|---|---|---|
| P1 | 0 | |
| P2, P5, P3, P6, P4 | 0.22 | 0 |

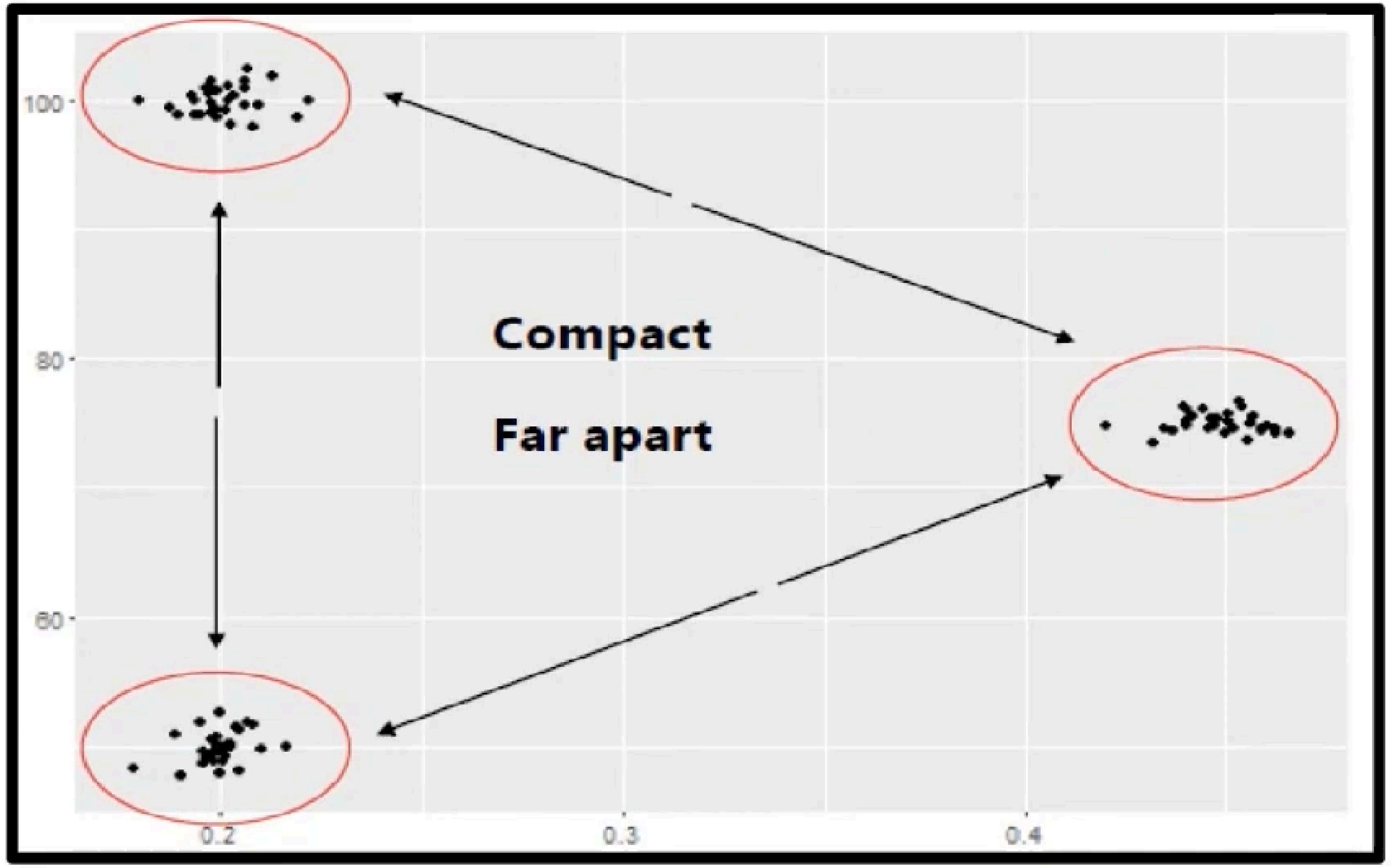


Repeat the exercise of combining the data points and updating the distance matrix, till all the data points are grouped to form one single cluster

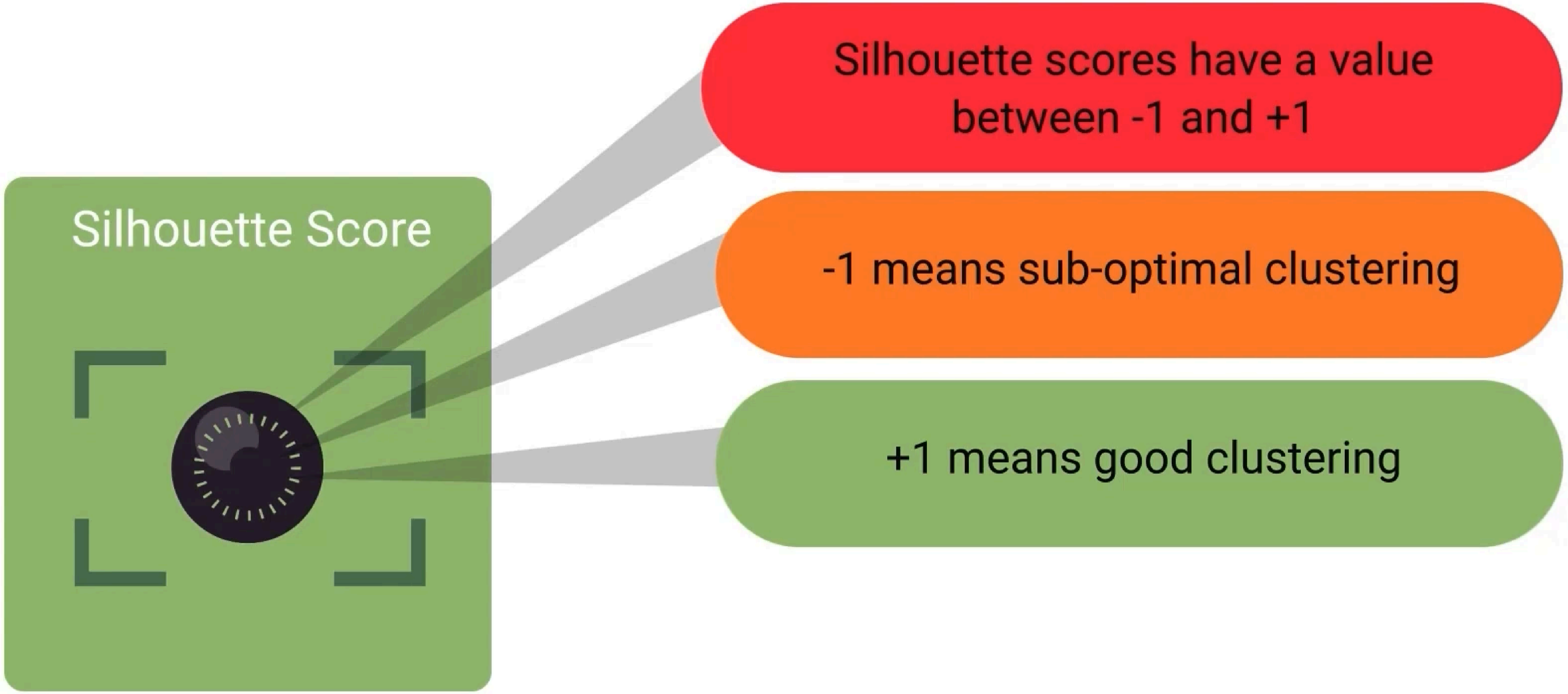



| | P1 | P2, P5, P3, P6, P4 |
|---|---|---|
| P1 | 0 | |
| P2, P5, P3, P6, P4 | 0.22 | 0 |

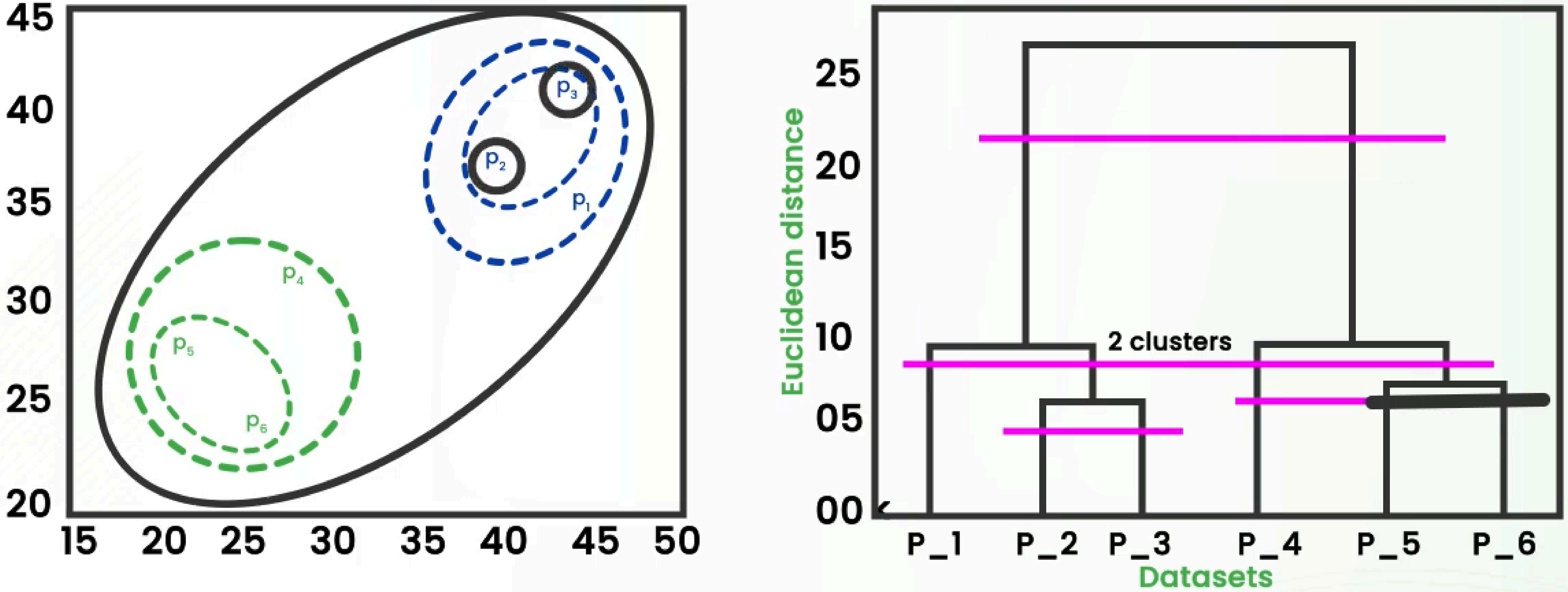





Silhouette scores can be used to check if the clusters obtained from the dendrogram are:


Silhouette Score
Silhouette scores have a value between -1 and +1
-1 means sub-optimal clustering
+1 means good clustering

- Agglomerative clustering is a type of hierarchi clustering
- Agglomerative clustering is a bottom-up approach wherein each data point is treated as a cluster, which are merged to form a successively large cluster
- Dendrogram is a pictorial representation to visualize the clusters
- Distance matrix is computed to find the two nearest data points



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Euclidean distance
2 clusters
P_1
P_2
P_3
P_4
P_5
P_6
Datasets